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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Regular Examinations August-2025

Common to ISE/CSE/CD/CY

SOFTWARE ENGINEERING WITH AGILE TECHNOLOGIES

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

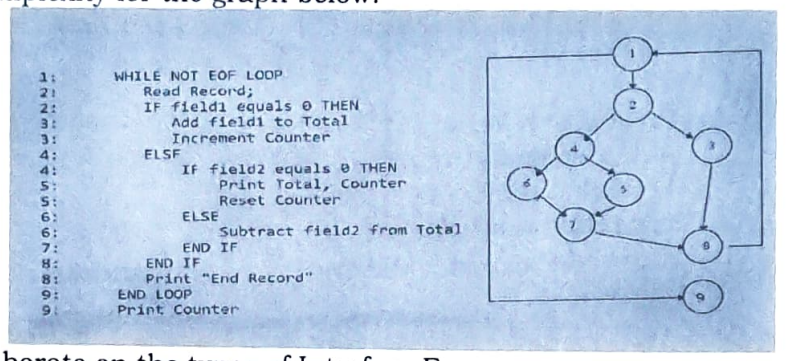
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	Classify the following requirement statements into Functional and Non-functional: a) The system must allow users to browse products and add them to a shopping cart. b) The app must allow users to rate and review products or services. c) The system should respond to user actions within 2 seconds. d) The system should prevent unauthorized access to sensitive information.	02	3	1
	1.2	Differentiate between Incremental and iterative development.	02	2	2
	1.3	List two purpose of Use case Diagrams.	02	2	3
	1.4	List any two advantages of Open Source Development.	02	1	2
	1.5	Give two differences between Alpha Testing and Beta Testing.	02	2	1
	1.6	Give two examples of legacy System Replacement	02	3	2
	1.7	Identify the type of risk for the given statements: a) Failure of a purchased component to perform as expected b) Uncontrolled changes to the project's scope after it has been initiated c) The size of the system has been underestimated. d) The underlying technology on which the system is built is superseded by new technology.	02	3	3
	1.8	Mention two advantages of Group Cohesiveness.	02	1	2
	1.9	List the principles of Kanban.	02	1	4
	1.10	List any four Agile development methods.	02	1	4

PART-B

2	a	Differentiate between Traditional Development and Agile Development Techniques.	04	2	4
	b	Considering the example of Online Banking System write any three Functional and Nonfunctional requirements for the same.	06	3	2
	c	With a diagram explain Boehm's Spiral Model.	06	2	1
3	a	Write a brief note on OpenSource Development with license models.	04	2	1
	b	Draw the sequence diagram for ATM operations.	06	3	3
	c	Draw use case diagram for a Library Management System for the cases of issue of books, searching books, updating the database.	06	3	3

		OR			
4	a	Elaborate the following Design pattern with examples: i) Façade pattern ii) Iterator Pattern	06	2	2
	b	Construct the following UML diagrams: i) State machine diagram for a microwave operation ii) Activity diagram for an Order Processing System	10	3	3
5	a	With a neat diagram explain the stages of Testing Process.	08	2	1
	b	Explain the stages of Development testing with suitable examples.	08	2	3
		OR			
6	a	Mention the steps of Path Testing. Calculate the Cyclomatic Complexity for the graph below:  	05	3	3
	b	Elaborate on the types of Interface Errors.	05	2	3
	c	Describe the benefits of Test-Driven development.	06	2	2
7	a	Discuss the factors that influence Project management.	06	2	3
	b	Elaborate the following risks with examples: i) Product risk ii) Project risk iii) Business risk	06	2	3
	c	Summarize Maslow's basic Hierarchy of needs.	04	2	3
		OR			
8	a	Elaborate the risk management process with a neat diagram.	08	2	3
	b	Discuss the factors that affect the group Communications.	08	2	4
9	a	Describe the Principles of Agile Methods.	10	2	4
	b	Summarize the Extreme Programming Release cycle with a neat diagram.	06	2	4
		OR			
10	a	Mention the advantages of Pair Programming.	04	2	4
	b	Discuss the terminologies of a Sprint cycle with a neat diagram.	06	2	4
	c	Summarize the role of Kanban in improving, measuring and managing Workflow.	06	2	4